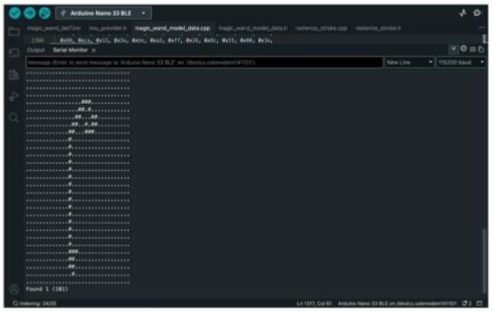
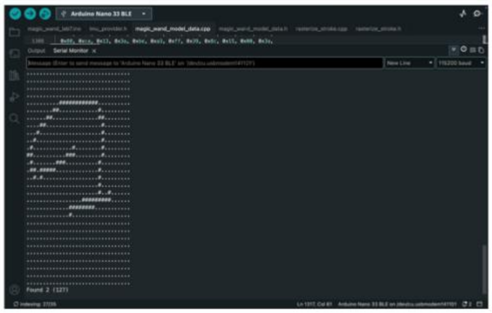
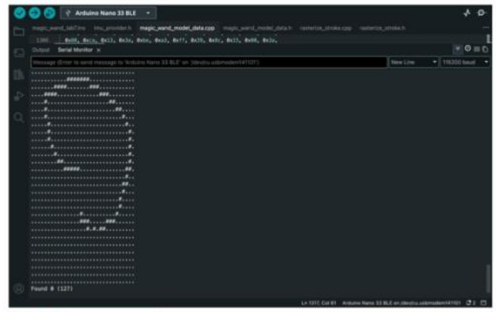
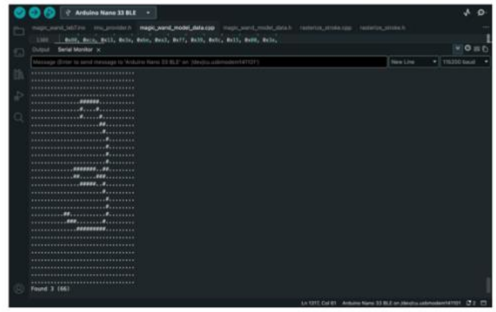
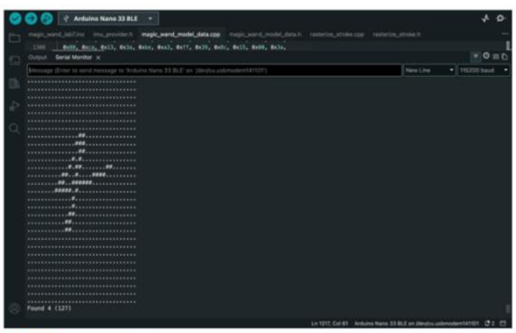
Baseline model

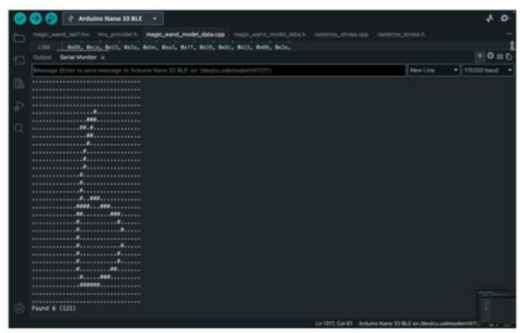


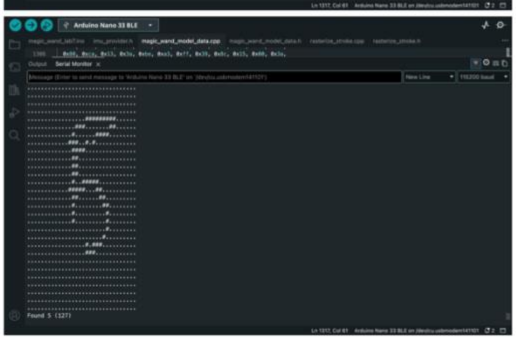


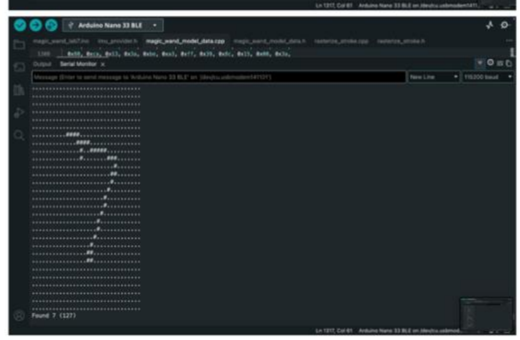


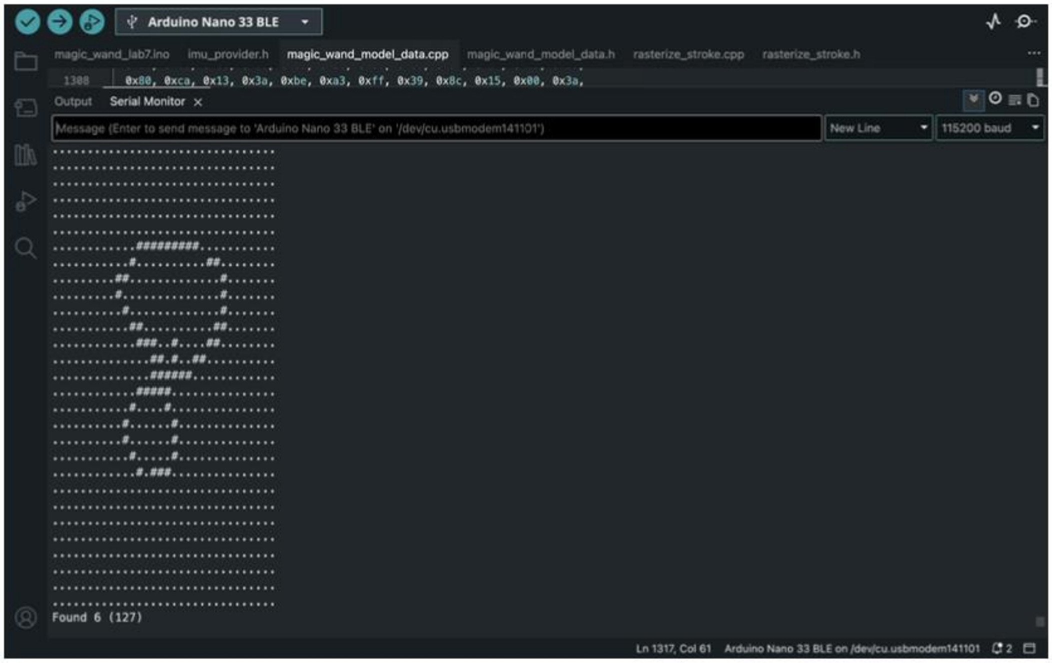


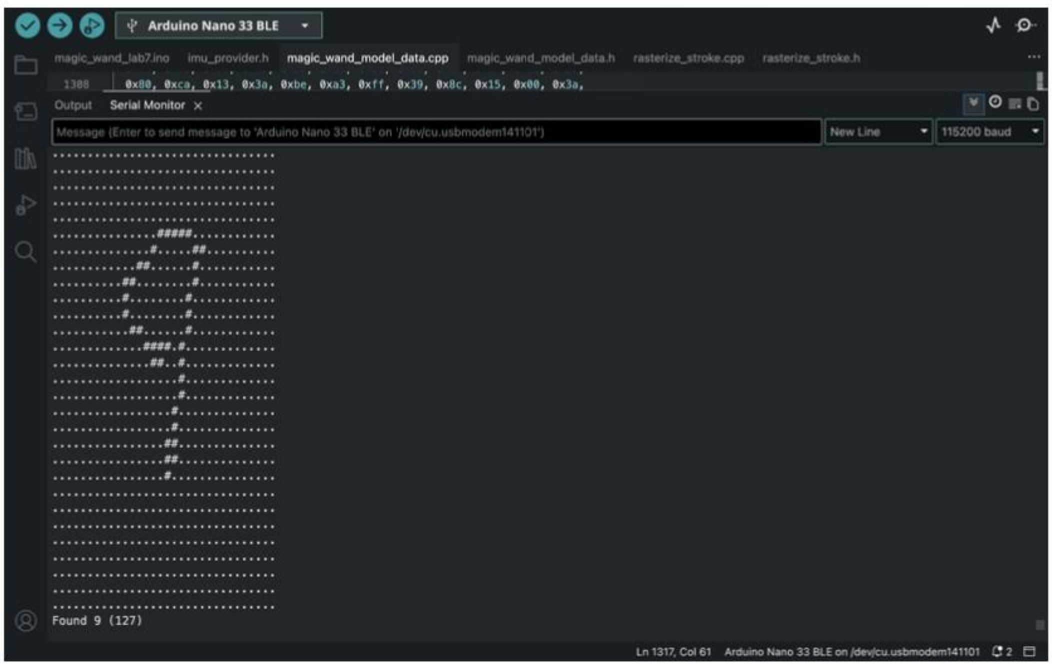




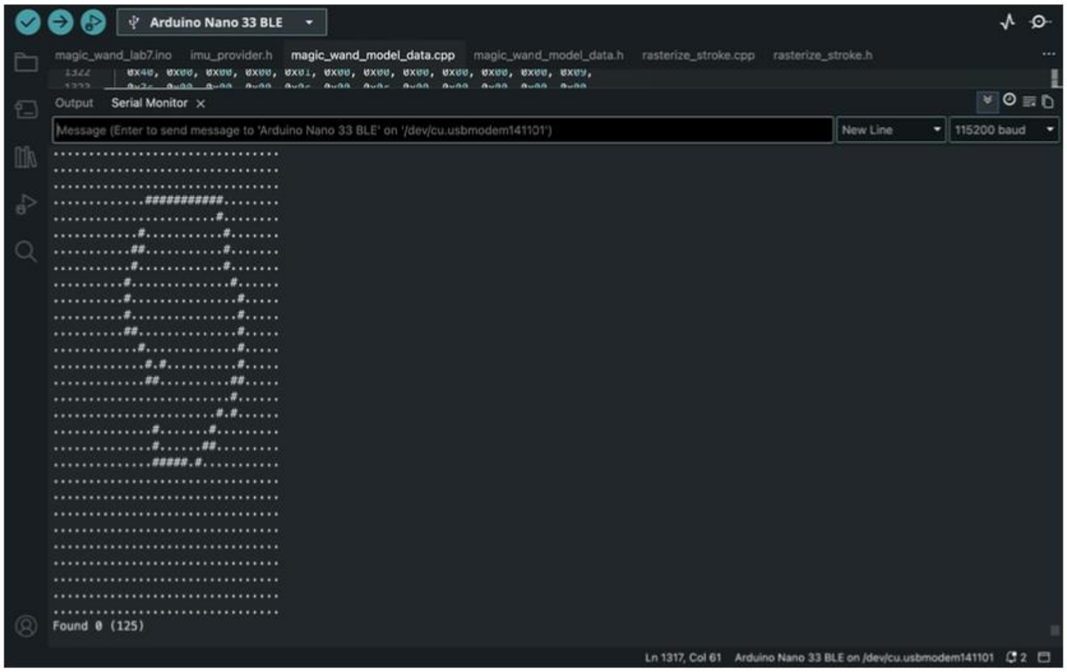


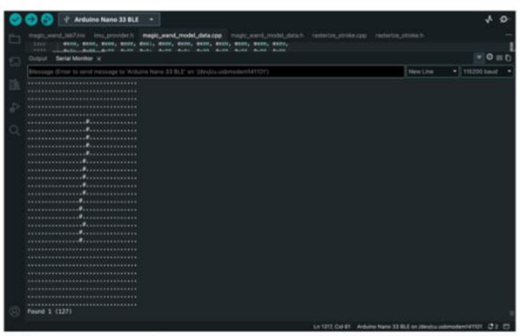


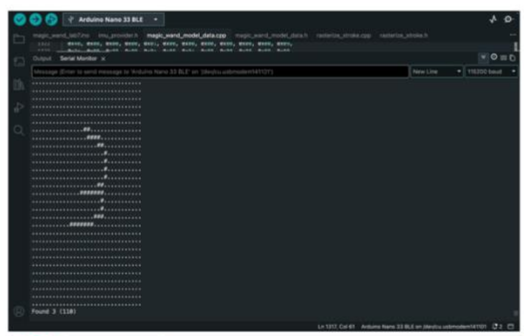

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 6 (127)
Ln 1317, Col 61 Arduino Nano 33 BLE on /dev/cu.usbmodem141101

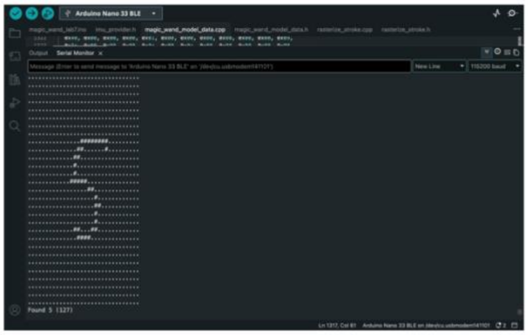

Arduino Nano 33 BLE
magic_wand_lab7.ino
imu_provider.h
magic_wand_model_data.cpp
magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 9 (127)
Ln 1317, Col 61 Arduino Nano 33 BLE on /dev/cu.usbmodem141101

Finetuned model








Arduino Nano 33 BLE
Output Serial Monitor
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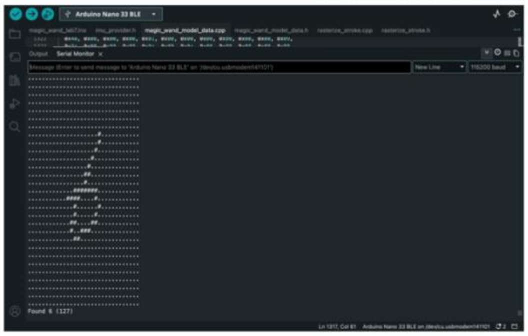
Arduino Nano 33 BLE
Output Serial Monitor
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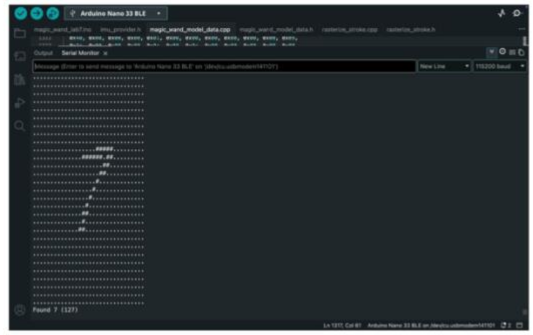
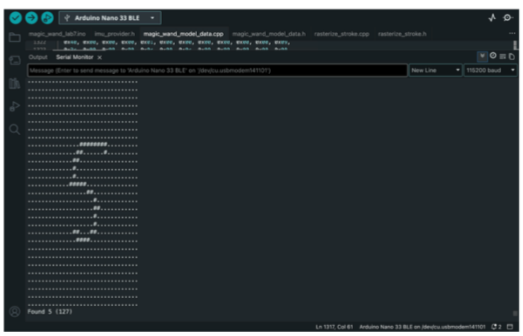
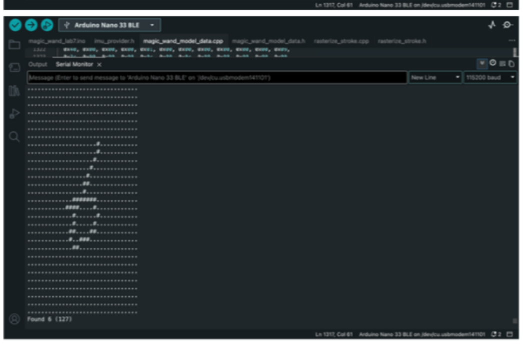
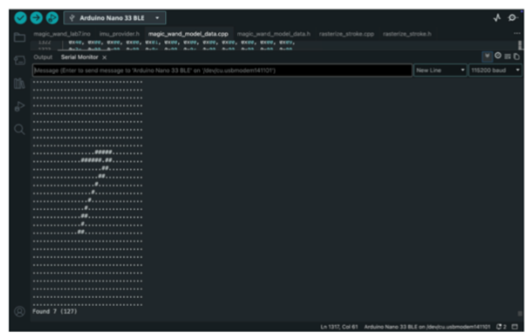

Arduino Nano 33 BLE
Output Serial Monitor
Found 7 (127)


Arduino Nano 33 BLE
Output Serial Monitor
Found 5 (127)


Arduino Nano 33 BLE
Output Serial Monitor
Found 6 (127)


Arduino Nano 33 BLE
Output Serial Monitor
Found 7 (127)


Arduino Nano 33 BLE
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magic_wand_model_data.h
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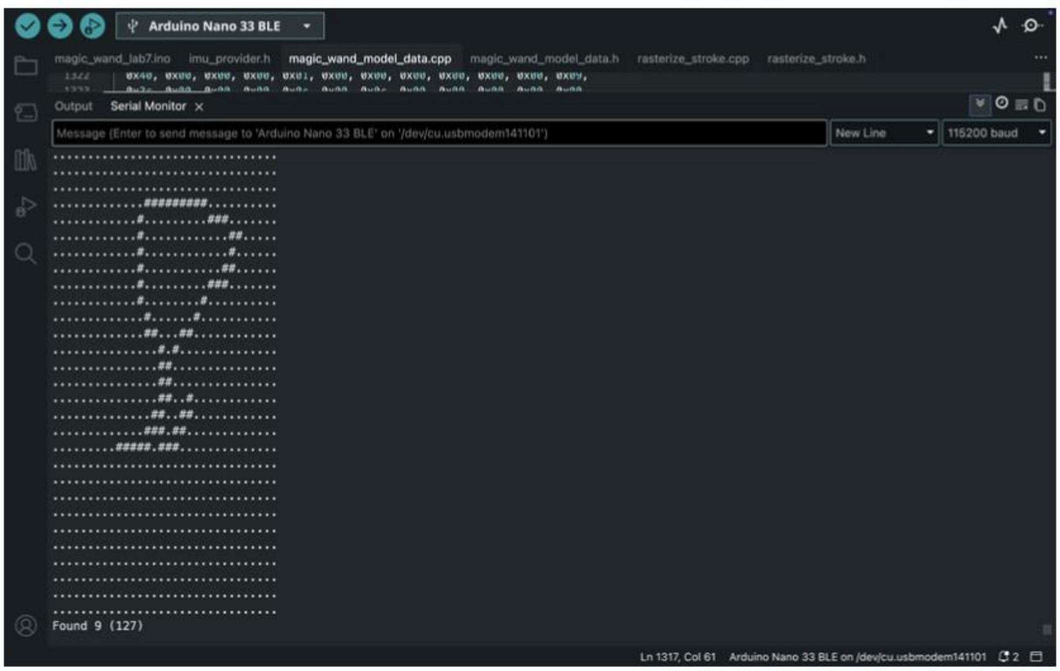
rasterize_stroke.h
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 2 (67)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101


Arduino Nano 33 BLE
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magic_wand_model_data.h
rasterize_stroke.cpp
rasterize_stroke.h
Output
Serial Monitor
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem141101')
New Line
115200 baud
Found 9 (127)
Ln 1317, Col 61
Arduino Nano 33 BLE on /dev/cu.usbmodem141101

Discussion:

Between the baseline and finetuned models, the baseline initially performed better. The main reason was poor dataset quality during finetuning.

After identifying these issues, we recollected and cleaned the data, fixed the labels, combined the JSON files, retrained the model, and redeployed it. The new dataset contained 100 balanced samples, with 10 examples per digit class.

This significantly improved performance. The updated model achieved 93% overall accuracy with a macro-average F1-score of approximately 0.93, showing that the earlier poor results were mainly caused by data quality issues rather than limitations of finetuning itself.

The new model performed especially well on digits 1, 2, and 5, which achieved perfect precision and recall. Digits 3, 4, 7, and 9 also showed strong performance with F1-scores around 0.90–0.95.

The weakest class was digit 8, with a recall of 0.70 and an F1-score of 0.78. Some 8s were misclassified as 0s or 6s because these digits share similar rounded motion patterns. Smaller errors also occurred between visually similar gestures such as 0 and 8, 3 and 7, and 9 and 4.

Overall, the second finetuning attempt was much more successful. The results show that finetuning performance is highly dependent on dataset quality. A small or mislabeled dataset can reduce performance, while a clean and balanced dataset can greatly improve accuracy. Further improvements would likely come from collecting more samples per class and training with more realistic deployment conditions to improve robustness.